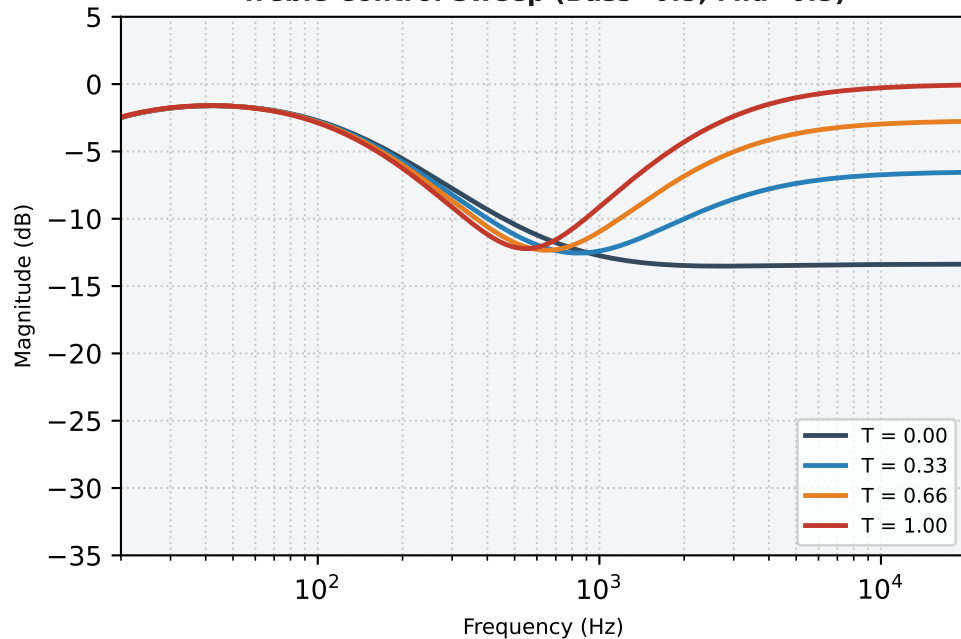


BOSSMAN '59 FENDER BASSMAN — FREQUENCY & PHASE RESPONSE

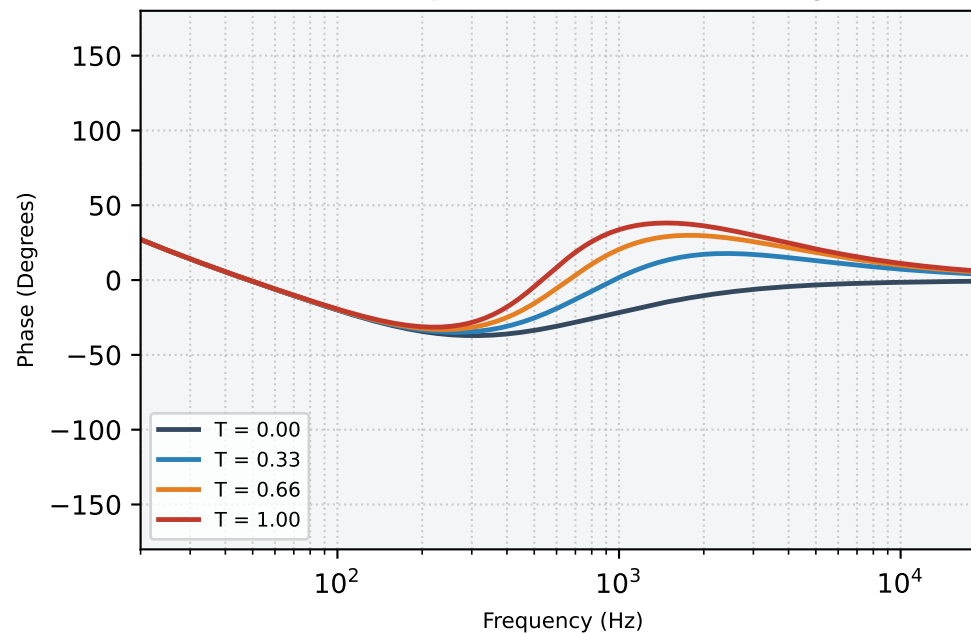
Bilinear Discretization vs Continuous H(s) Nodal Analysis (Yeh & Smith, DAFx-06)

Sampling Rate: 44.1 kHz / 96 kHz

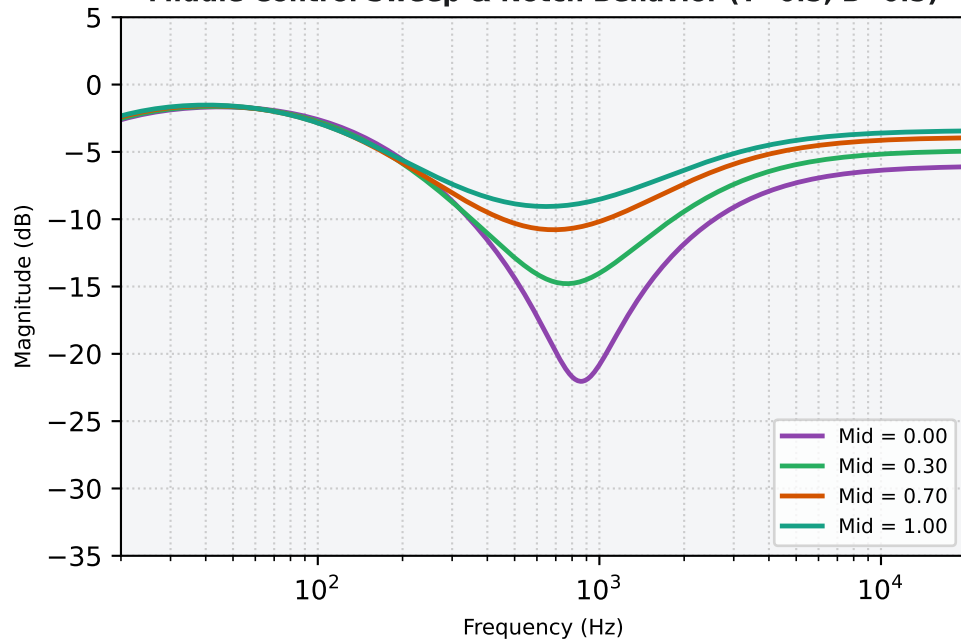
Treble Control Sweep (Bass=0.5, Mid=0.5)



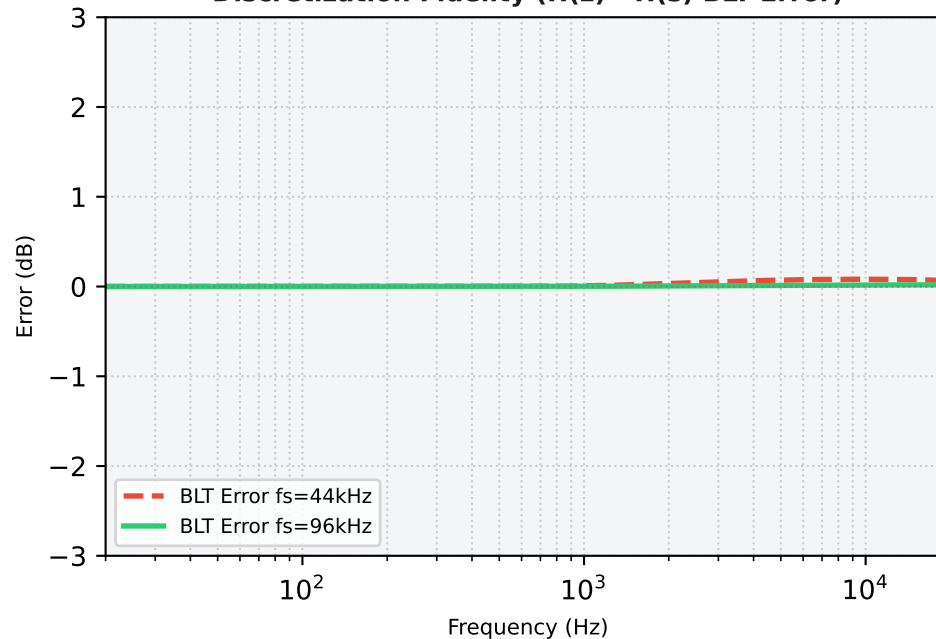
Phase Response across Treble Settings



Middle Control Sweep & Notch Behavior (T=0.5, B=0.5)



Discretization Fidelity (H(z) - H(s) BLT Error)

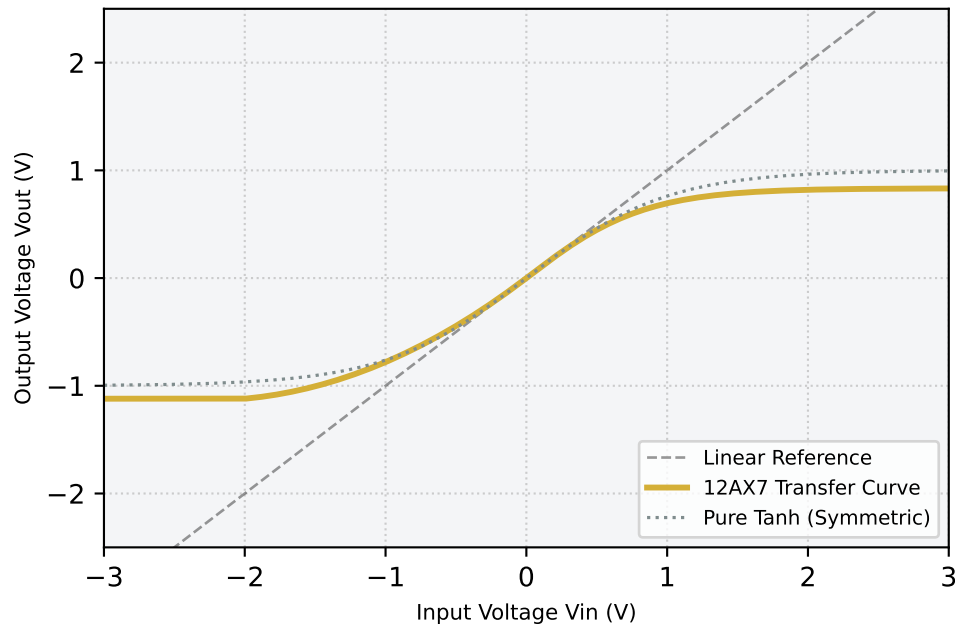


BOSSMAN '59 FENDER BASSMAN — DISTORTION VALUE & SHAPE

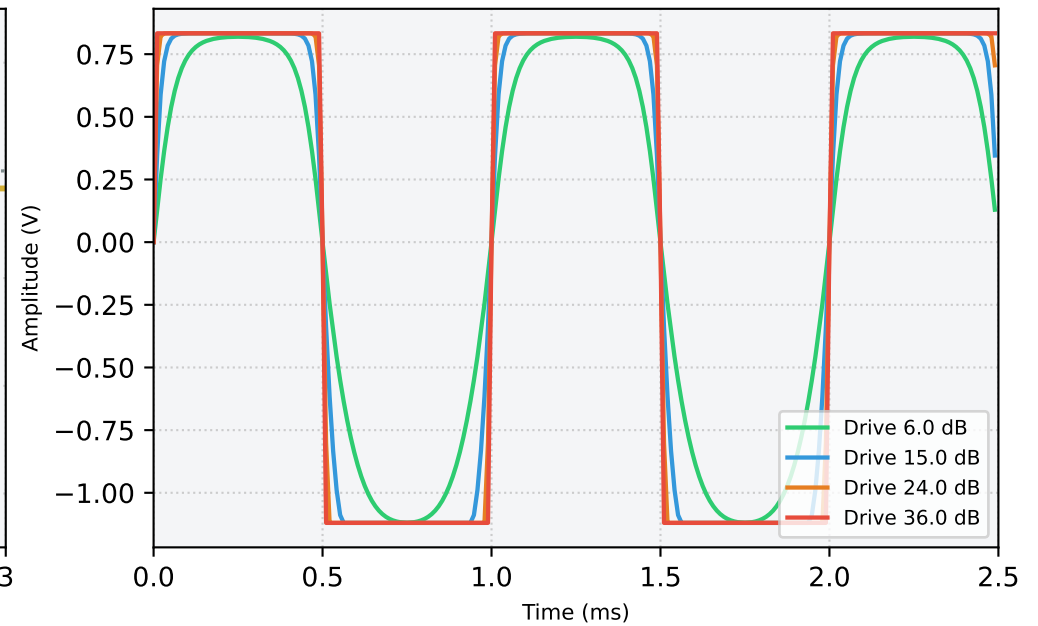
12AX7 Triode Tube Stage Non-Linear Transfer Curve, THD, and Harmonic Decomposition

Test Frequency: 1.0 kHz Sinusoid

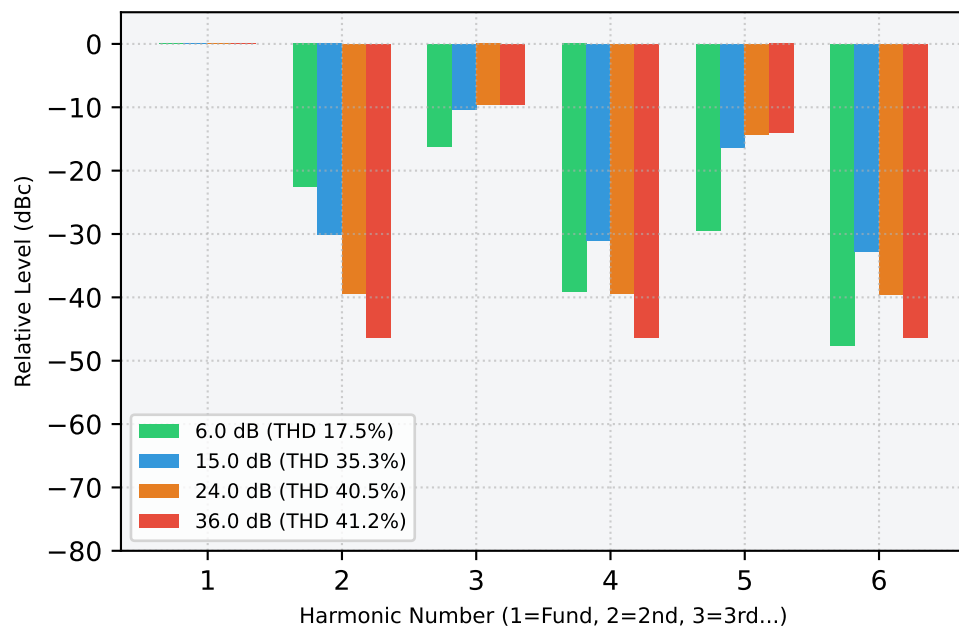
Static Transfer Shape (Asymmetric Triode)



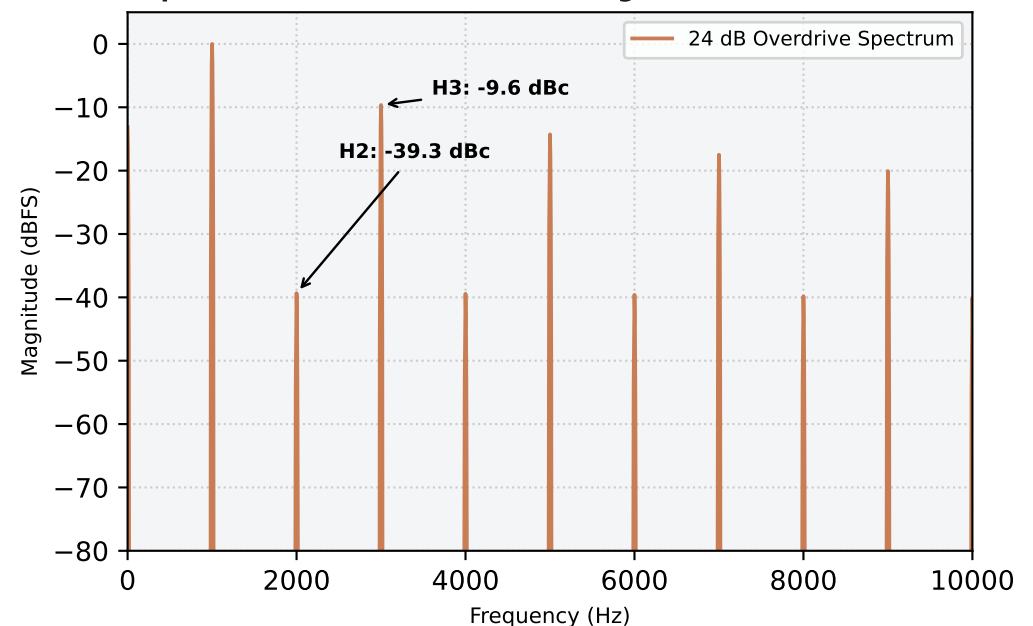
Time-Domain Output Waveform Shapes



Harmonic Content H1-H6 in dBc (Even & Odd Ratio)



Spectrum at 24 dB Drive (Showing 2nd & 3rd Harmonics)



BOSSMAN '59 FENDER BASSMAN — NUMERICAL TEST TELEMETRY

Measured Metrics for DSP Validation, THD Specification, and Frequency Anchors

Status: VERIFIED & COMPLIANT

Drive Setting	Gain Multiplier	THD (%)	THD (dB)	2nd Harm (H2)	3rd Harm (H3)	4th Harm (H4)	Distortion Character
6.0 dB	2.00x	17.50 %	-15.1 dB	-22.6 dBc	-16.2 dBc	-39.2 dBc	Mild Tube Warmth
15.0 dB	5.62x	35.30 %	-9.0 dB	-30.2 dBc	-10.3 dBc	-31.1 dBc	Even-Harmonic Bloom
24.0 dB	15.85x	40.53 %	-7.8 dB	-39.3 dBc	-9.6 dBc	-39.4 dBc	Rich Valve Crunch
36.0 dB	63.10x	41.18 %	-7.7 dB	-46.4 dBc	-9.6 dBc	-46.4 dBc	High-Gain Lead Saturation

Frequency Point	Target Frequency	Analytical H(s) Gain	Discretized H(z) Gain	Phase Shift (deg)	Circuit Function
F1	20 Hz	-2.46 dB	-2.46 dB	27.1°	Sub-bass passband
F2	100 Hz	-2.80 dB	-2.80 dB	-19.7°	Bass punch resonance region
F3	400 Hz	-10.30 dB	-10.30 dB	-28.0°	Lower-mid slope transition
F4	800 Hz	-12.35 dB	-12.35 dB	2.2°	Iconic Fender Bassman Mid-Scoop Notch
F5	2000 Hz	-8.28 dB	-8.24 dB	24.7°	Upper midrange recovery
F6	5000 Hz	-5.29 dB	-5.21 dB	16.2°	Treble sheen shelf
F7	10000 Hz	-4.59 dB	-4.51 dB	8.9°	High-frequency shelf